



Cemo CPH 320

Typical applications: Deep mine dewatering and public utility water transfer

CEMO Multistage Centrifugal Pumps are used for large volume water transfer in both deep mining dewatering applications including water transfer for public water utilities. Advantages include high mechanical and volumetric efficiencies, large flow rates and high heads. Materials of construction are sand and die cast PB1 Phosphor bronze (impellers, diffusers, shaft sleeves, neck rings, neck bushes, balance and counter disc components. All stage casings, suction and delivery covers are fabricated from W300 steel plate and stress relieved prior to final machining. Anti-abrasive coatings such as HVOF tungsten carbide, ceramic coatings and chrome plated coatings are used on wear susceptible wetted components or rotating element assemblies. All rotating element individual components and assemblies are balanced to ISO 1940/1 prior to assembly. All pump shafts are non-destructive tested and heat treated to relieve residual stresses. Pump bearings are white metal shell types.

All pumps are independently tested at The South African Bureau of Standards prior to delivery and a detailed Data Pack of material certification, non-destructive tests, balancing certification and verification of pump performance is documented and supplied on pump delivery to client.

All pumps and spares fully manufactured (machining and castings) at our Johannesburg SA works to ISO 9001:2015 requirements

Tel: +27 618-2003 E-Mail: sales@cemo.co.za

Physical Address: Crown Building, 64 Hanau Street, Jeppestown, JHB

Standard Trading Terms, Conditions and Warranties Apply And Are Available On request



Typical materials of construction

Suction cover:

W300 Steel plate fabrication with left or right hand flange orientation as per client's requirements

Delivery cover:

W300 Steel plate fabrication with vertical discharge flange configuration

Note: All flanges (rated to 12 Mpa) are to ANSI spec or drilled as per client's requirements

Stage casings:

300WA rolled and fabricated steel with Cast Iron or PB1/LG2 Bronze replaceable inserts

Binding bolts & nuts:

Certified 080-M40 (BS970-1991) steel threaded both ends

Shaft:

817-M40 or 709-M40 Condition "T" ultrasonically non-destructive tested and stress relieved for maximum tensile strength (Taper or Parallel drive end journal as per client's requirements)

Impellers / Diffusers / Guideports / Shaft Sleeves:

PB1 or LG2 Bronze, chill cast or 316 Stainless Steel as per client's requirements. If requested abrasive resistant HVOF Tungsten coating on all wear areas and tips

Neck Rings:

PB2 / LG4, chill cast and Tungsten coated to suit client's requirements

Balance Valve Assembly Wear Faces:

PB1 Bronze, Mild Steel or 316 Stainless Steel uncoated or hard faced i.e. Tungsten or Stellite Welded on faces

Bearing Housings:

Nodular Iron SG42

Balance Chamber:

W300 Fabricated steel

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Bearings:

Bronze shells with WM90 White Metal

Optional Equipment:

- Digital balance water return flow meter
- Pump monitoring kit (High and low pressure gauges including high pressure monitoring cock)
- Coupling – Grid or Gear type as per client's requirements
- Pump and motor baseplate assembly
- Wear Limit Indicator Switch – mechanical or proximity type to identify Wear Face deterioration
- Electrically actuated pump isolation valves and pump Non-Return valves

Pump protection: (compulsory supply with all pumps)

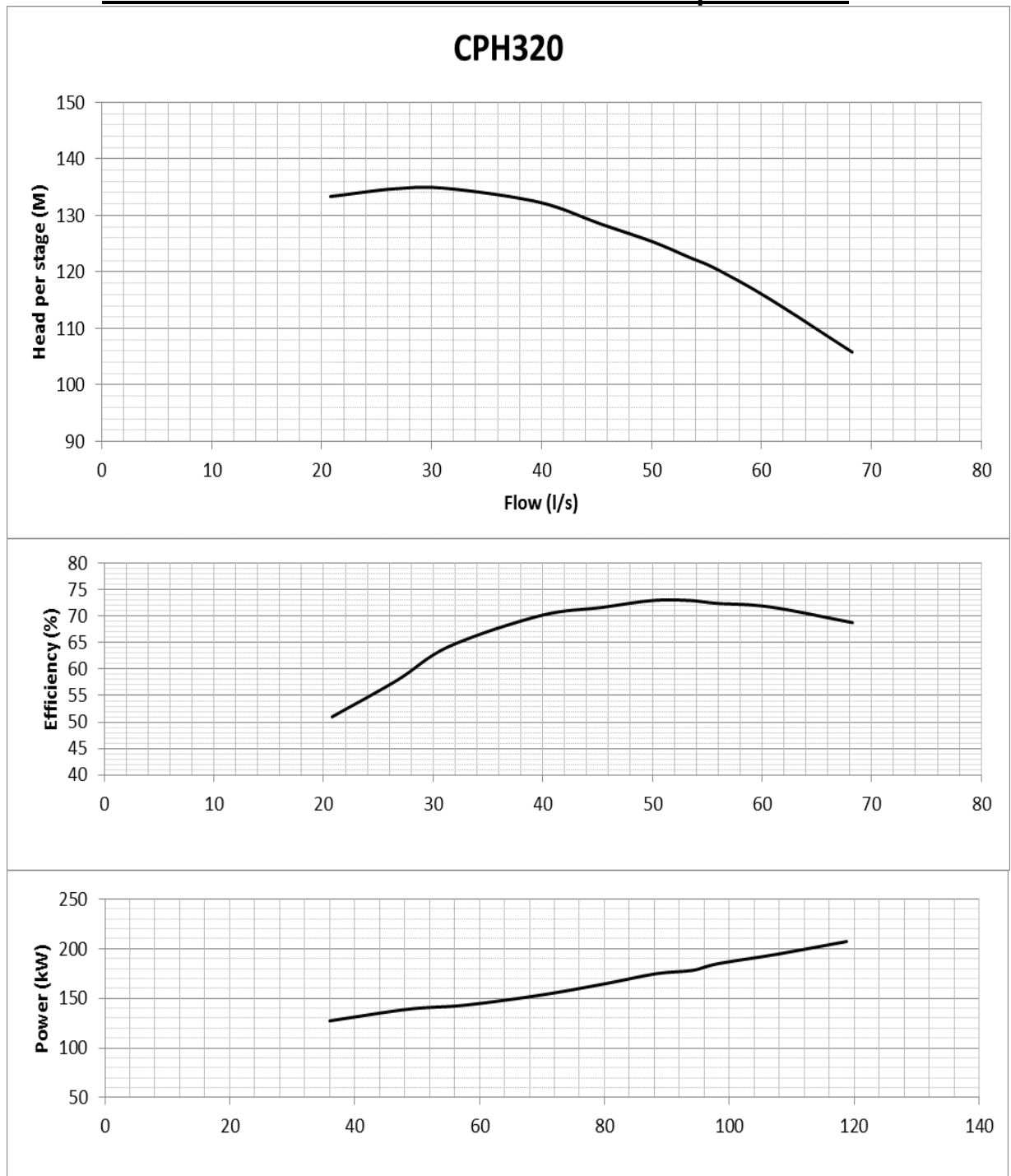
- For Balance disc assembly protection, fully adjustable wear limit switch with motor cut out sensors / pump motor trip function
- Bearing temperature probes
- Horizontal plane drive end bearing vibration monitoring probe

Services:

On site commissioning and monitoring both locally and globally by certified pump technicians with all on site monitoring equipment and telemetry



Cemo CPH 320 – Tested Pump Curve



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Pump speed	2980 rpm	Inlet diameter	150mm
Impeller Diameter	Ø335mm	Outlet diameter	125mm
NPSH (min)	10m	Q @ BEP	52 l/s